Wrong Centre, Wrong Vector, Wrong Scale Factor Answer Key

High School Geometry

Quick Answers

1. (a) horizontal component = 4, (b) vertical component = 3, —
2. (a) x-coordinate of B' = −6, (a) y-coordinate of B' = 15, —
3. (a) correct scale factor = 1.67, —
4. (a) x-coordinate of P' = 9, (a) y-coordinate of P' = 8, —
5. (a) correct linear scale factor = 1.5, —
6. (a) x-coordinate of N' = 14, (b) length of the correct image segment = 12, —

What is verified

10 of 16 answers machine-checked against SymPy, across 6 problems.
— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-CO.A – *problem 1*
HSG-SRT.A.1 – *problems 2, 4, 6*
HSG-SRT.A.2 – *problems 3, 5*
Difficulty 2–5 of 5 across 16 tagged checks.

Common wrong answers

1. *If they got −4: subtracted the image coordinate from the original instead of the other way round*
2. *If they got 1: added the scale factor to each coordinate instead of multiplying*
2. *If they got 8: added the scale factor to the second coordinate instead of multiplying*
3. *If they got 6: subtracted the two sides instead of dividing*
4. *If they got 10: dilated about the origin instead of the given centre*
4. *If they got 12: dilated the second coordinate about the origin instead of the given centre*
5. *If they got 2.25: used the area ratio itself as the linear scale factor*
6. *If they got 18: dilated about the origin instead of about the endpoint given as the centre*

1. Translation maps $A(3,-2)$ to $A'(7,1)$. Ana writes $(x,y) \to (x-4,\ y-3)$.

Solution (a): The vector runs *from* the original *to* the image, so each component is image minus original.

$$7 - 3$$

So the horizontal component is $\boxed{4}$

Solution (b): $1 - (-2) = 1 + 2$, so the vertical component is $\boxed{3}$

The correct rule is $(x, y) \to (x + 4,\ y + 3)$.

Solution (c) — instructor-judged. A model response: Ana subtracted in the wrong order, taking original minus image instead of image minus original. That flips the sign of both components, so her rule points the opposite way: applied to $A(3, -2)$ it gives $(-1, -5)$, which is nowhere near $A'(7, 1)$.

Full credit: states the image-minus-original order and says the reversal reverses the direction. *Half credit:* gives the correct order only. *No credit:* fixes the numbers with no account of the error.

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2. Dilation, centre the origin, factor 3, applied to $B(-2, 5)$. Ben reports $B'(1, 8)$.

Solution (a): A dilation about the origin *multiplies* each coordinate by the scale factor.

$$x' = 3(-2) = -6, \qquad y' = 3(5) = 15$$

So the coordinates are $\boxed{-6}$

and $\boxed{15}$

Solution (b) — instructor-judged. A model response: Ben added 3 to each coordinate instead of multiplying by 3. Adding a fixed amount to both coordinates is a *translation*, so what he actually performed was a slide of 3 right and 3 up. A slide leaves the figure exactly the same size, which is the opposite of what a dilation with factor 3 is meant to do.

Full credit: states the multiplying rule and names the move he actually made as a translation. *Half credit:* gives the rule without naming his move. *No credit:* redoes the arithmetic with no diagnosis.

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3. Similar triangles with sides 9 cm and 15 cm. Cara says the scale factor is 6.

Solution (a): A scale factor is a ratio of matching sides.

$$k = \frac{15}{9} = \frac{5}{3} = 1.6666\ldots$$

So $\boxed{k \approx 1.67}$

Solution (b) — instructor-judged. A model response: a scale factor answers "how many times as long", which is a division, not a subtraction. Cara's 6 is a difference of two lengths, so it is measured in centimetres — a scale factor has no unit at all, because the units cancel in the ratio. Her method would also give the same "factor" for a 1 cm side becoming 7 cm, which is clearly a much bigger enlargement than 9 cm to 15 cm.

Full credit: names the quotient rule and explains why a difference cannot be a scale factor. *Half credit:* gives the division without the reason. *No credit:* states the value alone.

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4. Dilation, centre $C(1, 4)$, factor 2, applied to $P(5, 6)$. A student reports $P'(10, 12)$.

Solution (a): Measure the offset from the centre, scale it, and add it back to the centre.

$$x' = 1 + 2(5 - 1) = 1 + 8 = 9$$
$$y' = 4 + 2(6 - 4) = 4 + 4 = 8$$

So the coordinates are $\boxed{9}$

and $\boxed{8}$

Solution (b) — instructor-judged. A model response: the centre of a dilation is the one point that does not move, and every other point moves directly away from it by the scale factor. Using the origin scales each point's offset *from the origin* instead of from $C(1, 4)$, so the image comes out the right size but in the wrong place. Here the student's $P'(10, 12)$ is twice as far from the origin as P; the correct $P'(9, 8)$ is twice as far from C.

Full credit: says the centre is fixed and explains what using the origin scales instead. *Half credit:* states the centre form of the rule only. *No credit:* recomputes with no diagnosis.

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5. Similar rectangles of area 32 cm^2 and 72 cm^2. A student says the linear scale factor is 2.25.

Solution (a): Areas of similar figures scale by k^2, so the area ratio *is* k^2.

$$k^2 = \frac{72}{32} = 2.25 \qquad k = \sqrt{2.25}$$

So $\boxed{k = 1.5}$

Solution (b) — instructor-judged. A model response: when every length is multiplied by k, both dimensions of a rectangle are stretched, so the area is multiplied by k^2. The ratio $72 \div 32$ is therefore already the square of the factor, and a square root is needed before it can be used as a length ratio. The student's 2.25 would only be the linear factor if area scaled the same way length does, which it does not. Checking: a 4×8 rectangle scaled by 1.5 becomes 6×12, whose area is 72 ✓

Full credit: explains why the given ratio is k^2 and names the square-root step. *Half credit:* names the square root without the reason. *No credit:* states the value alone.

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6. $\overline{MN}$ with $M(2, 3)$, $N(6, 3)$ dilated by 3 about M. A student used the origin and got $M'(6, 9)$, $N'(18, 9)$.

Solution (a): With M as the centre, M stays put and N's offset from M is tripled.

$$x' = 2 + 3(6 - 2) = 2 + 12$$

So the correct x-coordinate of N' is $\boxed{14}$

Solution (b): The correct image runs from $M'(2, 3)$ to $N'(14, 3)$, so its length is $|14 - 2|$.

$\boxed{12}$

Solution (c) — instructor-judged. A model response: a dilation multiplies every length by the scale factor no matter which centre is chosen, so both images are $3 \times 4 = 12$ units long — the student's segment from $(6, 9)$ to $(18, 9)$ measures 12 as well. What the centre decides is *where* the image lands. Dilating about the origin moved M to $(6, 9)$, but the problem specified M as the centre, so M was supposed to stay exactly where it was.

Full credit: says lengths are preserved under any centre and identifies position as what the centre controls. *Half credit:* notes the lengths agree with no account of the centre's role. *No credit:* calls the student's answer wholly right or wholly wrong.